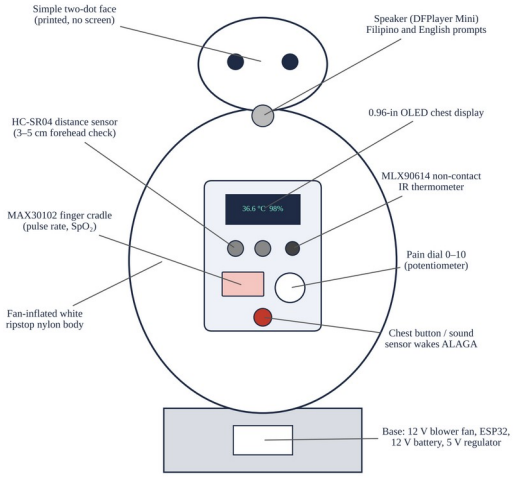
ALAGA

A Low-Cost Soft Inflatable Healthcare Companion Robot with Non-Contact Temperature, Pulse Oximetry, and Pain-Scale Triage for Evacuation Centers



Front view, not to scale. Height about 1.1 m when inflated.

ROBOTICS AND INTELLIGENT MACHINES – TEAM (RIM-T)

RESEARCH MANUSCRIPT

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ALAGA: A Low-Cost Soft Inflatable Healthcare Companion Robot with Non-Contact Temperature, Pulse Oximetry, and Pain-Scale Triage for Evacuation Centers

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Abstract

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Evacuation centers in Albay fill quickly when Mayon Volcano is restless, and a few health workers must check many evacuees for fever, low oxygen, and pain. Healthcare robots on the market cost far more than a barangay can pay. Inspired by the soft, inflatable healthcare companion of the animated film Big Hero 6, this study built ALAGA, a low-cost robot with a fan-inflated ripstop nylon body that houses an ESP32 controller, a non-contact infrared thermometer, a pulse oximeter sensor, a distance sensor, a 0–10 pain dial, a chest display, and a speaker that gives instructions in Filipino and English. A state-machine program wakes the robot when a person presses its chest or calls out, guides a short screening, and flags readings outside set limits for a health worker. With ____ consenting volunteers, ALAGA's readings were compared with a clinical forehead thermometer and a reference pulse oximeter. Temperature had a mean bias of ____ °C (95% limits of agreement ____ to ____ °C), pulse rate ____ bpm, and SpO_2 ____% (Arms = ____%). Flagging had a sensitivity of ____% and a specificity of ____%. One screening took ____ s. Users gave a System Usability Scale score of ____ and rated the soft body ____ approachable than a rigid shell (p = ____). The robot cost PHP ____ to build. A low-cost soft robot can share the routine screening load in crowded evacuation centers.

Keywords: healthcare companion robot; soft robotics; inflatable robot; pulse oximetry; non-contact thermometer; evacuation center

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INTRODUCTION

Background of the Study

Mayon Volcano was raised to Alert Level 3 in January 2026, and residents of the 6-kilometer Permanent Danger Zone were evacuated (DZRH, 2026). In early May 2026, ash fell on 87 villages in three towns, and more than 300 families moved to evacuation centers (Khaosod English, 2026). Evacuation centers are crowded, and fine volcanic ash irritates the eyes and airways, so coughs, fevers, and breathing complaints rise during long stays (Horwell & Baxter, 2006).

A barangay health worker must check each evacuee, often hundreds in a day. The basic checks are body temperature, pulse rate, oxygen saturation (SpO_2), and pain. A pulse oximeter finds low blood oxygen before a person looks ill (Jubran, 2015), and a 0–10 numeric pain scale is quick and reliable for adults (Hawker et al., 2011). These checks are simple, but doing them for every evacuee takes time that health workers need for the sick.

Robots that help with care are known as socially assistive robots (Feil-Seifer & Matarić, 2005). Older adults accept them more when they look friendly and are easy to use (Broadbent et al., 2009). The animated film Big Hero 6 (Hall & Williams, 2014) showed a soft, inflatable healthcare companion that scans a person, asks for a pain score from 0 to 10, and cannot hurt anyone because its body is air. Real inflatable robot arms have been built for safe contact with people (Sanan et al., 2011), and soft robots are safer around people than rigid ones (Rus & Tolley, 2015). This study asks whether such a companion can be built from low-cost parts for an evacuation center in Albay.

Research Gap and Rationale

Vital-sign kiosks and healthcare robots exist, but they are rigid, costly, and not made for crowded shelters. Low-cost sensors such as the MAX30102 and MLX90614 are popular in student projects, yet their readings are rarely checked against reference instruments with proper agreement statistics (Bland & Altman, 1986). No low-cost robot found by the researchers combines a soft inflatable body, non-contact temperature, pulse oximetry, pain-scale triage, and spoken instructions in Filipino. It is also not known whether a soft body makes people, especially children and older adults, more willing to be screened than a rigid machine.

Objectives of the Study

This study aims to design, build, and evaluate ALAGA, a low-cost soft inflatable healthcare companion robot that screens temperature, pulse rate, SpO_2, and pain and flags evacuees who need a health worker. Specifically, it seeks to answer the following questions:

1. How closely do ALAGA's temperature, pulse rate, and SpO_2 readings agree with a reference clinical forehead thermometer and a reference fingertip pulse oximeter (mean bias, 95% limits of agreement, mean absolute error, and Arms)?
2. How accurately does ALAGA flag readings outside the screening limits (sensitivity, specificity, and accuracy)?
3. How long does ALAGA take to complete one screening, from wake-up to spoken result?
4. How do users rate the usability of ALAGA, and is there a significant difference in perceived approachability between its soft inflatable body and a rigid shell?
5. How much does ALAGA cost to build and to run for one day?

Hypotheses

The following null hypotheses were tested at the 0.05 level of significance:

- $H_0 1$: There is no significant difference between ALAGA's readings and the reference readings for temperature, pulse rate, and SpO_2.
- $H_0 2$: There is no significant difference in perceived approachability ratings between the soft inflatable body and the rigid shell.

Significance of the Study

The results of this study will be useful to the following:

Evacuees. Children and older adults can be screened by a friendly, soft robot without waiting in a long line.

Barangay health workers and nurses. Routine checks are shared with the robot, so more time goes to evacuees flagged YELLOW or RED.

LGUs and the PDRRMO. A screening aid that costs a few thousand pesos can be placed in every evacuation center.

Schools used as shelters. The same robot can serve the school clinic between disasters.

Future researchers. A tested, open design and an agreement method for low-cost health sensors.

The study supports SDG 3 (good health and well-being), SDG 9 (industry, innovation, and infrastructure), and SDG 11.5 (fewer people affected by disasters).

Scope and Delimitation

This study covers the following:

- One stationary prototype, about 1.1 m tall when inflated, with a fan-inflated ripstop nylon body.
- Screening of forehead temperature, pulse rate, SpO_2, and self-reported pain (0–10).
- Spoken prompts in Filipino and English, a chest display, and a log stored in the robot's memory.
- Tests with healthy volunteers aged 15 years and above in a school setting, after IRB approval and informed consent.
- A bench test of the infrared thermometer with a temperature-controlled water-bath target.

The study does not cover the following:

- Walking, arms, or lifting; ALAGA does not move or touch the person.
- Diagnosis or treatment; ALAGA only screens and refers. It is not a registered medical device.
- Blood pressure, respiratory rate, and ECG.
- Real patients with low SpO_2 or fever; abnormal values are covered by the water-bath bench test and by logic test cases.
- Use in an actual evacuation center, which is left for a pilot study.

Definition of Terms

ALAGA. From the Filipino word for care; the robot built in this study.

Healthcare companion robot. A robot that keeps a person company while it checks basic health signs and gives simple advice.

Soft inflatable body. A sewn fabric shell kept full of air by a small fan, so it gives way when touched.

Non-contact infrared thermometer. A sensor that reads skin temperature from the infrared light the skin gives off (MLX90614).

Pulse oximetry. Measuring SpO_2 and pulse rate from red and infrared light passed through a fingertip (MAX30102).

SpO_2. The percentage of hemoglobin carrying oxygen in arterial blood.

Numeric Rating Scale (NRS). A 0–10 pain score where 0 is no pain and 10 is the worst pain imaginable.

Mean bias and limits of agreement. The average difference between ALAGA and the reference, and the range that holds 95% of the differences.

Arms. Root-mean-square error of SpO_2 against the reference; ISO 80601-2-61 accepts 4% or less.

Flag. A YELLOW or RED triage result that tells the person to see a health worker.

System Usability Scale (SUS). A 10-item questionnaire scored from 0 to 100 (Brooke, 1996).

FRAMEWORK

Review of Related Literature

Soft robots are made of materials that bend and give way, which makes them safer around people than rigid machines (Rus & Tolley, 2015). Inflatable links are light and absorb impact; Sanan et al. (2011) built an inflatable arm so that a robot could touch and help people safely. A body kept full of air by a fan is the simplest form of this idea and costs little to make.

Socially assistive robots help people through talk and company rather than physical work (Feil-Seifer & Matarić, 2005). Older adults accept healthcare robots more when the robot is useful, easy to use, and not frightening (Broadbent et al., 2009). Faces that look almost human can feel eerie, an effect called the uncanny valley (Mori et al., 2012), so a simple two-dot face was chosen.

Pulse oximeters shine red and infrared light through a fingertip and compare how much of each is absorbed with each heartbeat (Allen, 2007; Jubran, 2015). The ISO 80601-2-61 standard requires an Arms of 4% or less for SpO_2 from 70% to 100% (International Organization for Standardization [ISO], 2017b). Infrared thermometers estimate temperature from the infrared energy the skin gives off (Fraden, 2016); forehead skin is cooler than the body core, so a calibration offset is needed and clinical thermometers are tested under ISO 80601-2-56 (ISO, 2017a). The 0–10 numeric pain scale is valid and quick for adults (Hawker et al., 2011).

When a new instrument is compared with a reference, correlation is not enough; Bland and Altman (1986) recommend the mean difference and its 95% limits of agreement. Usability can be measured with the 10-item System Usability Scale (Brooke, 1996).

Review of Related Studies

Table 1

Related Studies and Their Relation to the Present Study

Study	What was done	Relation to the present study
Sanan et al. (2011)	Built an inflatable robot arm for safe contact with people.	Supports an air-filled body for ALAGA.
Broadbent et al. (2009)	Reviewed what makes older adults accept healthcare robots.	Guides the friendly look and simple use.
Jubran (2015)	Reviewed how pulse oximeters work and where they fail.	Sets SpO_2 limits and known error sources.
Bland & Altman (1986)	Proposed limits of agreement for comparing two methods.	Method used to test ALAGA's readings.

Note. Table created by the researchers, 2026.

Synthesis

The literature shows that soft bodies are safer and friendlier, that pulse oximetry and infrared thermometry can screen for hypoxemia and fever, and that new instruments must be checked against a reference with agreement statistics. No study found by the researchers joins these into one low-cost robot for evacuation centers or tests whether a soft body makes people more willing to be screened.

Theoretical Framework

The study rests on four ideas, each answering a gap (Table 2). Socially assistive robotics explains how a robot can share routine work with health workers. Soft robotics and the uncanny valley guide the body and face. Photoplethysmography and infrared thermometry explain how the sensors work, and agreement analysis tests whether their readings can be trusted. A state machine with screening limits turns the readings into a simple GREEN, YELLOW, or RED result.

Table 2

Research Gaps and the Theories or Concepts Used to Address Them

Research gap	Description	Theory or concept	How it is used
Too few health workers	Hundreds of evacuees need routine checks (Khaosod English, 2026).	Socially assistive robotics	Robot does routine screening; humans treat.
Machines can scare people	Rigid or almost-human robots can feel unsafe or eerie.	Soft robotics; uncanny valley	Air-filled body and a simple two-dot face.
Cheap sensors may be wrong	Low-cost modules are rarely validated.	PPG, IR thermometry, agreement analysis	Calibrate, then test with Bland–Altman.
Who needs help first	Raw numbers mean little to evacuees.	State machine with screening limits	GREEN, YELLOW, or RED with spoken advice.

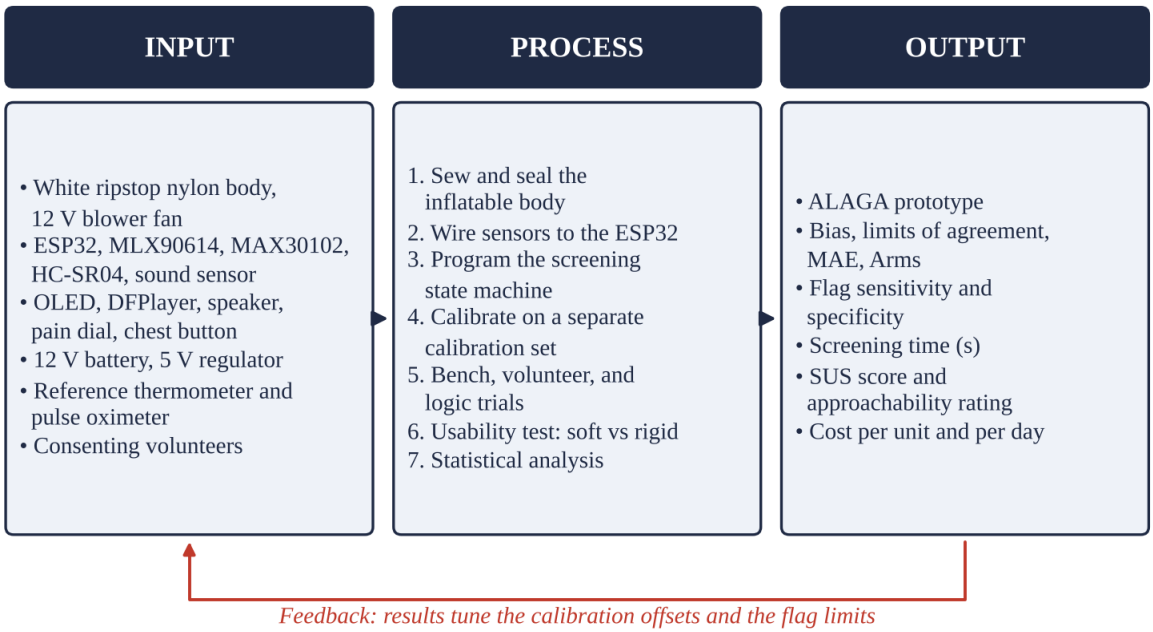
Note. Table created by the researchers, 2026.

Conceptual Framework

The study follows the input–process–output model shown in Figure 1. The inputs are the materials, instruments, and volunteers; the process covers construction, calibration, testing, and analysis; and the outputs answer each objective. Results are fed back to tune the calibration offsets and the flag limits.

Figure 1

Conceptual Framework of the Study in Input–Process–Output Form



Note. Figure created by the researchers, 2026.

METHODOLOGY

Research Design

This is a quantitative developmental and method-comparison study with a within-subjects usability test. The robot is built and calibrated first; then its readings are compared with reference instruments on volunteers, its flag logic is tested, and users rate it with a soft body and with a rigid shell in counterbalanced order. Every value comes from the researchers' own tests.

Table 3

Variables of the Study

Independent (varied)	Dependent (measured)	Controlled (held fixed)
Measurement method (ALAGA; reference instrument)	Bias, limits of agreement, MAE, Arms	Room temperature 26–30 °C; 5-min rest
Water-bath target temperature (°C)	Sensitivity, specificity, accuracy Screening time (s)	Forehead distance 3–5 cm; left index finger
Body type (soft inflatable; rigid shell)	SUS score; approachability rating (1–5)	Same prompts, order alternated; same robot shape and size

Note. Table created by the researchers, 2026.

Materials and Equipment

The components were bought from local electronics shops and online stores. The reference clinical forehead thermometer and fingertip pulse oximeter were borrowed from the school clinic. Expenses are listed in Appendix A.

Table 4

Materials and Components of ALAGA

Material / Component	Quantity	Purpose
White 70D ripstop nylon, seam tape, zipper	3 m; 1 set	Inflatable body and head
12 V DC blower fan with finger guard	1	Keeps the body inflated
ESP32 development board	1	Controller
MLX90614 (GY-906) infrared thermometer	1	Forehead temperature
MAX30102 pulse oximeter module	1	Pulse rate and SpO_2
HC-SR04 ultrasonic sensor	1	Forehead distance check
0.96-in OLED display (SSD1306)	1	Chest display
DFPlayer Mini, microSD card, 3 W speaker	1 set	Spoken prompts
KY-037 sound sensor, push button, 10 kΩ dial	1 each	Wake-up and pain score

12 V 7 Ah SLA battery, LM2596 regulator, fuse	1 set	Power
IRF520 MOSFET module, wires, perfboard	1 set	Fan switch and wiring
Reference forehead thermometer and pulse oximeter	1 each	Reference readings (borrowed)

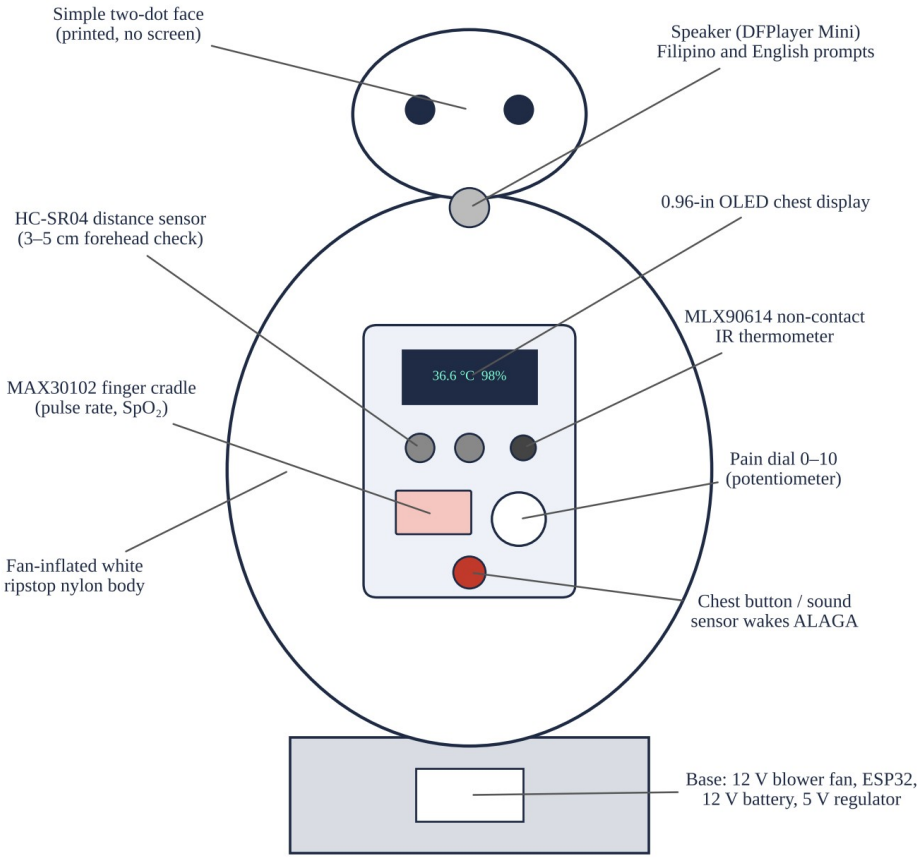
Note. Table created by the researchers, 2026.

Specification of the Device

- Body: white 70D ripstop nylon, sewn and seam-taped, about 1.1 m tall; kept inflated by a 12 V blower fan in the base (Figure 2).
- Sensor panel on the chest: MLX90614 infrared thermometer and HC-SR04 distance sensor at forehead height, MAX30102 in a padded finger cradle, pain dial, chest button, and 0.96-in OLED display.
- Voice: DFPlayer Mini with a 3 W speaker; recorded prompts in Filipino and English.
- Wake-up: chest button or a KY-037 sound sensor that responds to a loud call such as “Aray!”.
- Controller and power: ESP32 with a state-machine program; 12 V 7 Ah sealed lead-acid battery and a 5 V buck regulator; log saved to the ESP32's flash memory as a CSV file.

Figure 2

Design and Parts of ALAGA



Front view, not to scale. Height about 1.1 m when inflated.

Note. Figure created by the researchers, 2026.

Construction and Preparation

Inflatable Body

The body was cut from ripstop nylon in two front and two back panels, sewn with flat seams, and sealed with seam tape. The head was a separate sewn shell joined at the neck. A zippered opening at the chest holds the sensor panel, and the base collar is clamped around the fan outlet with a hose clamp. A small vent lets air leak out so the fan keeps the body firm without bursting it.

Electronics and Sensors

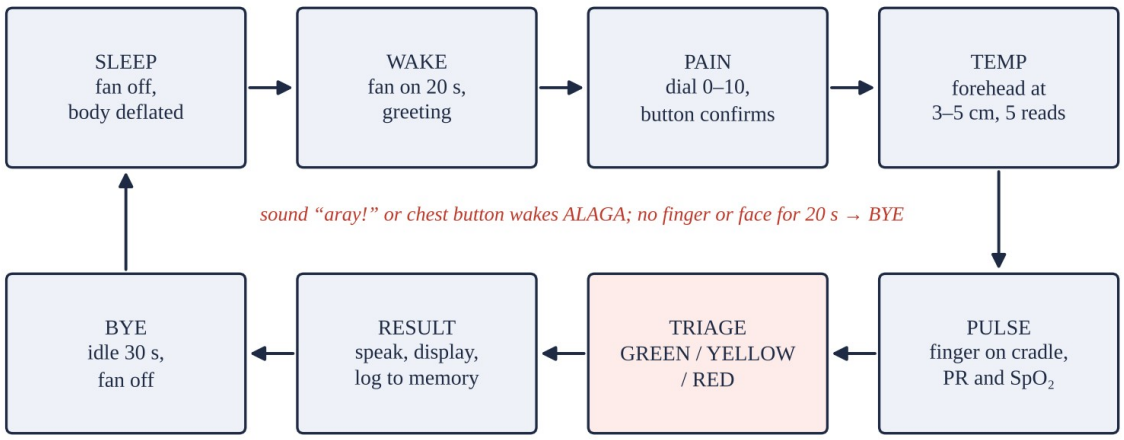
The MLX90614, MAX30102, and OLED share one I²C bus at 100 kHz. The HC-SR04 echo line passes through a 1 kΩ/2 kΩ divider to protect the 3.3 V ESP32. The fan is switched by an IRF520 MOSFET module with a flyback diode. Pin connections are in Appendix G.

Control Logic

The program follows the state machine in Figure 3. In SLEEP the fan is off and the body lies flat. A press on the chest or a loud call starts the fan, and after 20 s ALAGA greets the person and asks for a pain score. It then reads the forehead temperature five times when the distance sensor shows 3 to 5 cm, reads pulse rate and SpO_2 when a finger is on the cradle, applies the limits in Table 5, speaks the result, shows it on the display, and saves it. After 30 s with no one present, the fan stops.

Figure 3

Screening State Machine of ALAGA



Note. Figure created by the researchers, 2026.

Table 5

Screening Limits and Triage Levels

Measure	GREEN (normal)	YELLOW (see health worker)	RED (see health worker now)
Temperature (°C)	below 37.5	37.5 to 38.9	39.0 or higher
SpO_2 (%)	95 to 100	92 to 94	below 92
Pulse rate at rest (bpm)	60 to 100	40–59 or 101–130	below 40 or above 130
Pain (0–10 NRS)	0 to 3	4 to 6	7 to 10

Note. Table created by the researchers, 2026.

Calibration

Forehead skin temperature from the MLX90614 was converted to body temperature with a linear equation fitted to readings from the first 10 volunteers (calibration set). Constant offsets for pulse rate and SpO_2 were found from the same set. These volunteers were not used again in the agreement analysis.

Experimental Setup

All tests were done indoors at a room temperature of 26 to 30 °C, away from fans, air conditioners, and direct sunlight. Volunteers sat and rested for 5 minutes before screening, removed nail polish from the left index finger, and wiped sweat from the forehead. ALAGA and the reference instrument were used one after the other within 1 minute, in alternating order.

Table 6

Trials Conducted in the Study

Trial set	What is measured	Number	Reference
Bench temperature	IR reading of a black target in a water bath, 35.0–40.0 °C	6 targets × 10 readings	Digital probe thermometer
Calibration set	Temperature, pulse rate, SpO_2	10 volunteers × 3	Clinic instruments
Validation set	Temperature, pulse rate, SpO_2, time	____ volunteers × 3	Clinic instruments
Logic test	Triage of set input values around each limit	60 cases	Table 5 limits
Usability	SUS; approachability, soft vs rigid	____ volunteers	Counterbalanced order

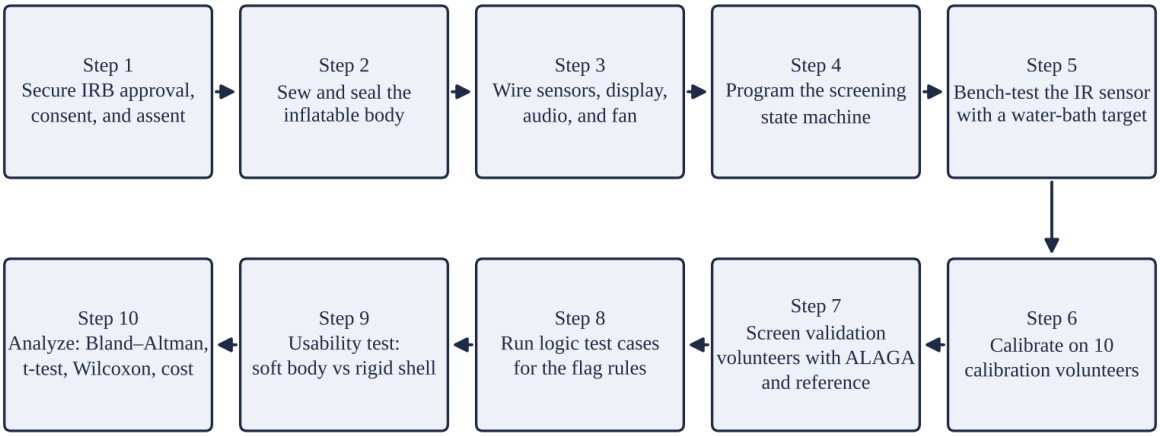
Note. Table created by the researchers, 2026.

Data Gathering Procedure

The ten steps in Figure 4 were followed. Each volunteer received a code number, and no names were written on the data sheets (Appendix B).

Figure 4

Flowchart of the Research Procedure



Note. Figure created by the researchers, 2026.

Statistical Treatment

The following formulas were used:

$d = \text{ALAGA reading} - \text{reference reading}$	where difference for each paired reading	(1)
$\text{Bias} = \Sigma d \div n;\ \text{LoA} = \text{Bias} \pm 1.96 \times SD(d)$	where mean bias and 95% limits of agreement	(2)
$\text{MAE} = \Sigma\lvert d\rvert \div n;\ \text{Arms} = \sqrt{(\Sigma d^2 \div n)}$	where mean absolute error; root-mean-square error	(3)
$\text{Sensitivity} = TP \div (TP + FN);\ \text{Specificity} = TN \div (TN + FP)$	where flags correctly raised; normal results correctly passed	(4)
$\text{Accuracy} = (TP + TN) \div N$	where overall correct triage decisions	(5)
$t_s = t_{result} - t_{wake}$	where screening time (s)	(6)
$\text{SUS} = 2.5 \times [\Sigma(\text{odd item} - 1) + \Sigma(5 - \text{even item})]$	where System Usability Scale score (0–100)	(7)

Agreement was shown with Bland–Altman plots. A paired t-test tested $H_0 1$ for each measure. Flag decisions were summarized in a confusion matrix. Approachability ratings (1–5) for the soft body and the rigid shell were compared with the Wilcoxon signed-rank test ($H_0 2$). The level of significance was 0.05, and Microsoft Excel and JASP were used.

Safety and Ethical Considerations

This study involves human participants, so it followed the ISEF rules for human participants research used by the NSTF. Before any testing:

- The research plan was reviewed and approved by the school's Institutional Review Board (IRB).
- Each volunteer signed an informed consent form; volunteers below 18 years also gave assent, with written consent from a parent or guardian (Appendix E).
- Procedures were non-invasive: a forehead reading from 3 to 5 cm and a fingertip clip. No blood, drugs, or exercise were involved, and anyone could stop at any time.
- Volunteers with readings outside the limits were referred to the school nurse.
- Data were coded, kept on a password-protected computer, and deleted after the fair.
- All sources are cited in APA 7th edition; no data were fabricated.

Electrical and physical risks were controlled as listed in Appendix D: a finger guard on the fan, a fuse on the battery line, low-voltage wiring inside a closed box, wiping of the sensors with 70% alcohol between volunteers, and no small parts within reach.

FINDINGS

Results and Discussion

[NEED: every ____ in this chapter must be replaced with values the researchers measured. No value may be entered that was not measured.]

Results for Objective 1: Agreement With the Reference Instruments

Table 7 shows the infrared thermometer against the water-bath target. Across the targets, the mean error was ____ °C after calibration.

Table 7

Bench Test of the Infrared Thermometer Against a Water-Bath Target

Target (°C)	Probe reading (°C)	ALAGA mean ± SD (°C)	Error (°C)
35.0	____	____ ± ____	____
36.0	____	____ ± ____	____
37.0	____	____ ± ____	____
37.5	____	____ ± ____	____
38.0	____	____ ± ____	____
40.0	____	____ ± ____	____

Note. Table created by the researchers, 2026.

Table 8 shows the agreement of ALAGA with the reference instruments in the ____ validation volunteers. Temperature had a bias of ____ °C with limits of agreement from ____ to ____ °C. Pulse rate had a bias of ____ bpm, and SpO_2 had a bias of ____% with an Arms of ____%, which ____ the 4% limit of ISO 80601-2-61. The paired t-test gave p = ____, ____, and ____, so $H_0 1$ is ____.

Table 8

Agreement of ALAGA With the Reference Instruments (Validation Set)

Measure	n	Bias	95% LoA	MAE	Arms	p
Temperature (°C)	____	____	____ to ____	____	____	____
Pulse rate (bpm)	____	____	____ to ____	____	____	____
SpO_2 (%)	____	____	____ to ____	____	____	____

Note. Table created by the researchers, 2026.

Figure 5

Bland–Altman Plots of Temperature, Pulse Rate, and SpO_2

[NEED: Bland–Altman plots from the measured validation data: difference (ALAGA − reference) against the mean of the two, with lines for the bias and the 95% limits of agreement.]



Note. Figure created by the researchers, 2026.

Results for Objective 2: Accuracy of the Flags

Table 9 shows the triage decisions against the decisions made from the reference readings, including the logic test cases. Sensitivity was ____%, specificity ____%, and accuracy ____%.

Table 9

Confusion Matrix of ALAGA Flags

	Reference: flag	**Reference: normal**	**Total**
ALAGA: flag	TP = ____	FP = ____	____
ALAGA: normal	FN = ____	TN = ____	____
Total	____	____	____

Note. Table created by the researchers, 2026.

Results for Objective 3: Screening Time

One screening took a mean of ____ s (SD ____ s). The longest step was ____ (Table 10).

Table 10

Time Taken per Screening Step

Step	**Mean (s)**	**SD (s)**
Wake-up and inflation	____	____
Greeting and pain score	____	____
Temperature	____	____
Pulse rate and SpO_2	____	____
Result and advice	____	____
Total	____	____

Note. Table created by the researchers, 2026.

Results for Objective 4: Usability and Approachability

The mean SUS score was ____ (SD ____). Approachability was rated ____ for the soft body and ____ for the rigid shell (Wilcoxon z = ____, p = ____), so $H_0 2$ is ____ (Table 11).

Table 11

Usability and Approachability Ratings

Measure	**Soft body**	**Rigid shell**	**Test**	**p**
SUS score (0–100)	____	—	Mean ± SD	—
Approachability (1–5), median	____	____	Wilcoxon z = ____	____
Willing to be screened again (%)	____	____	—	—

Note. Table created by the researchers, 2026.

Results for Objective 5: Cost

ALAGA cost PHP ____ to build and PHP ____ per day to run (Table 12).

Table 12

Cost of Building and Running ALAGA

Item	Amount
Parts for one unit (Appendix A)	PHP ____
Energy per 12-hour day (Wh)	____
Electricity per day at PHP ____/kWh	PHP ____
Screenings per battery charge	____

Note. Table created by the researchers, 2026.

Summary of Results by Objective

Table 13

Summary of Results by Objective

Objective	Result	Decision
1. Agreement	Bias ____ °C, ____ bpm, ____ %; Arms ____%	$H_0$1 ____
2. Flag accuracy	Sensitivity ____%, specificity ____%	—
3. Screening time	____ s per person	—
4. Usability	SUS ____; soft vs rigid p = ____	$H_0$2 ____
5. Cost	PHP ____ per unit; PHP ____ per day	—

Note. Table created by the researchers, 2026.

CONCLUSIONS

Assessment of the Findings

Based on the results of the study, the following conclusions were drawn:

1. ALAGA was built with a fan-inflated soft body, an infrared thermometer, a pulse oximeter sensor, a pain dial, and spoken prompts, all controlled by an ESP32.
2. Its temperature, pulse rate, and SpO_2 readings had biases of ____ °C, ____ bpm, and ____%, with an SpO_2 Arms of ____%; $H_0$1 is ____.
3. Its flags reached ____% sensitivity and ____% specificity.
4. One screening took ____ s on average.
5. Users gave a SUS score of ____ and rated the soft body ____ approachable than the rigid shell (p = ____); $H_0$2 is ____.
6. One unit cost PHP ____ to build and PHP ____ per day to run.

Limitations

Only healthy volunteers were screened, so few real abnormal readings were seen; fever was covered by a water-bath target and low SpO_2 only by logic test cases. Low-cost pulse oximeter sensors can read higher in people with darker skin and with cold or moving fingers. The robot was tested in a school, not in a real evacuation center, and was not tested for weeks of continuous use.

Applications

Evacuation centers, barangay health stations, and school clinics in Albay can use ALAGA at the entrance to screen people while health workers attend to those it flags. Between disasters, the same unit can check students in the school clinic.

RECOMMENDATIONS

Based on the findings and limitations of the study, the researchers recommend the following:

1. Run a pilot in an evacuation center with the Albay PDRRMO and the Provincial Health Office.
2. Replace the MAX30102 with a medical-grade oximeter module and test across skin tones.
3. Add blood pressure and respiratory rate, and send flagged results to the health worker's phone.
4. Add voice prompts in Bikol and a simple speech-recognition pain score for people who cannot turn the dial.
5. Test the fabric and fan for dust and ash, and add a washable cover.

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APPENDICES

Appendix A. Materials, Quantity, and Expenses

Prices are estimates from local and online stores in October 2026. Replace them with the amounts on the official receipts.

Table 14

Expense Summary of the Study

Item	Qty	Estimated unit price (PHP)	Subtotal (PHP)
Ripstop nylon 70D, white (per meter)	3	150	450
Seam tape, zipper, thread, hose clamp	1 lot	200	200
12 V DC blower fan with guard	1	250	250
ESP32 development board	1	380	380
MLX90614 (GY-906) module	1	550	550
MAX30102 module	1	180	180
HC-SR04 ultrasonic sensor	1	70	70
0.96-in OLED display	1	180	180
DFPlayer Mini with 8 GB microSD	1	300	300
3 W 4 Ω speaker	1	80	80
KY-037 sound sensor, button, 10 kΩ dial	1 set	160	160
12 V 7 Ah SLA battery	1	650	650
LM2596 regulator, IRF520 module, fuse	1 set	180	180
Wires, perfboard, plywood base, box	1 lot	550	550
Rigid shell for the usability test	1	300	300
70% alcohol, wipes, data sheets	1 lot	150	150
Total (estimated)			**4,630**

Note. Table created by the researchers, 2026.

Appendix B. Raw Data Sheets

Table 15

Raw Data Sheet for Volunteer Screenings

Code	Trial	ALAGA T (°C)	Ref T (°C)	ALAGA PR	Ref PR	ALAGA SpO_2	Ref SpO_2	Pain	Time (s)
V01	1								
V01	2								
V01	3								
V02	1								
V02	2								
V02	3								

Note. Table created by the researchers, 2026.

Appendix C. Photographs of the Procedure

[NEED: dated photographs of each step in Figure 4, each with a bold label and the note "Photographs taken by the researchers, 2026." No faces of volunteers without written permission.]

Appendix D. Risk Register

Table 16

Risk Register of the Study

Hazard	Risk	Control
Blower fan	Finger caught in the blades	Metal finger guard; fan inside the closed base
Battery and wiring	Short circuit, heat	Inline fuse; low-voltage wiring in a closed box; charging only under supervision
Shared sensors	Spread of germs	Wipe with 70% alcohol between volunteers
Infrared and LED light	Eye discomfort	Sensors aimed at the forehead and fingertip only
Wrong reading	Missed fever or low SpO_2	Reference reading taken for every volunteer; referral to the school nurse
Personal data	Loss of privacy	Code numbers; password-protected files; deleted after the fair

Note. Table created by the researchers, 2026.

Appendix E. Ethics Statement and Informed Consent

This study involved human participants. The research plan, consent form, and data sheets were approved by the school Institutional Review Board on [date] before any volunteer was screened. Participation was voluntary, and volunteers could stop at any time without penalty.

Informed consent (summary read to each volunteer): “You are invited to test ALAGA, a robot that checks temperature, pulse, and oxygen level. A sensor will read your forehead from a short distance, and you will place one finger in a sensor. The same will be done with the school clinic's instruments. It takes about 5 minutes and does not hurt. Your name will not be recorded. If a reading is outside the normal range, we will ask the school nurse to check you.”

[NEED: signed consent forms, parental consent and minor assent forms, and the IRB approval form.]

Appendix F. ISEF Forms

[NEED: signed Checklist for Adult Sponsor (1), Student Checklist (1A), Research Plan/Project Summary, Approval Form (1B), Risk Assessment Form (3), Human Participants Form (4) with IRB approval, and informed consent forms.]

Appendix G. Wiring Diagram and Program

Table 17

Pin Connections of ALAGA

Part	Part pin	ESP32 pin	Note
MLX90614, MAX30102, OLED	SDA / SCL	GPIO 21 / GPIO 22	Shared I²C bus, 100 kHz, 3.3 V
HC-SR04	TRIG / ECHO	GPIO 32 / GPIO 33	ECHO through 1 kΩ/2 kΩ divider; 5 V supply
Pain dial (10 kΩ)	Wiper	GPIO 34	Ends to 3.3 V and GND
Chest button	One leg	GPIO 27	Other leg to GND; internal pull-up
KY-037 sound sensor	D0	GPIO 26	Threshold set with the onboard trimmer
DFPlayer Mini	TX / RX	GPIO 16 / GPIO 17	1 kΩ resistor on RX; speaker on SPK1/SPK2
IRF520 module	SIG	GPIO 25	Switches the 12 V blower fan
LM2596 regulator	OUT +5 V	VIN (5 V)	Input from the 12 V battery through a fuse

Note. Table created by the researchers, 2026.

The full program (alaga.ino, Arduino IDE, ESP32 Dev Module) is in the project folder firmware/alaga. It uses the Adafruit MLX90614, SparkFun MAX3010x, Adafruit SSD1306, and DFRobotDFPlayerMini libraries. Audio prompts are saved on the microSD card as /mp3/0001.mp3 to /mp3/0010.mp3: greeting, pain question, forehead prompt, finger prompt, GREEN, YELLOW, RED, goodbye, “come closer”, and “try again”.

[NEED: photograph or drawing of the final wiring.]

Appendix H. Curriculum Vitae

[NEED: one-page curriculum vitae of each researcher with a 2 × 2 photo.]